NLP

16-06-26

LSTM:

2) Input Gate:

$$i_t = \sigma(z_t) = \sigma[w_i \times [h_{t-1} + x_t] + b_i]$$

- w_i: learnable parameter for input gate
- b_i: bias for input gate

- input gate learns what relationship needs to be captured between h_{t-1} and x_t
- it also learns importance of current token x_t for given sequence.

using f_t and i_t we compute C_t – the cell state

$$\tilde{C}_t = \tanh[w_c \times [h_{t-1} + x_t] + b_c]$$

$$C_t = (f_t \times C_{t-1}) + (i_t \times \tilde{C}_t)$$

3) Output Gate:

$$O_t = \sigma[w_o \times [h_{t-1} + x_t] + b_o]$$

$$h_t = O_t \times \tanh(C_t)$$

- output gate learns how much information computed in given timestamp we need pass forward as output

Summary of LSTM operations:

$$f_t = \sigma[w_f \times [h_{t-1} + x_t] + b_f]$$

$$i_t = \sigma[w_i \times [h_{t-1} + x_t] + b_i]$$

$$\tilde{C}_t = \tanh[w_c \times [h_{t-1} + x_t] + b_c]$$

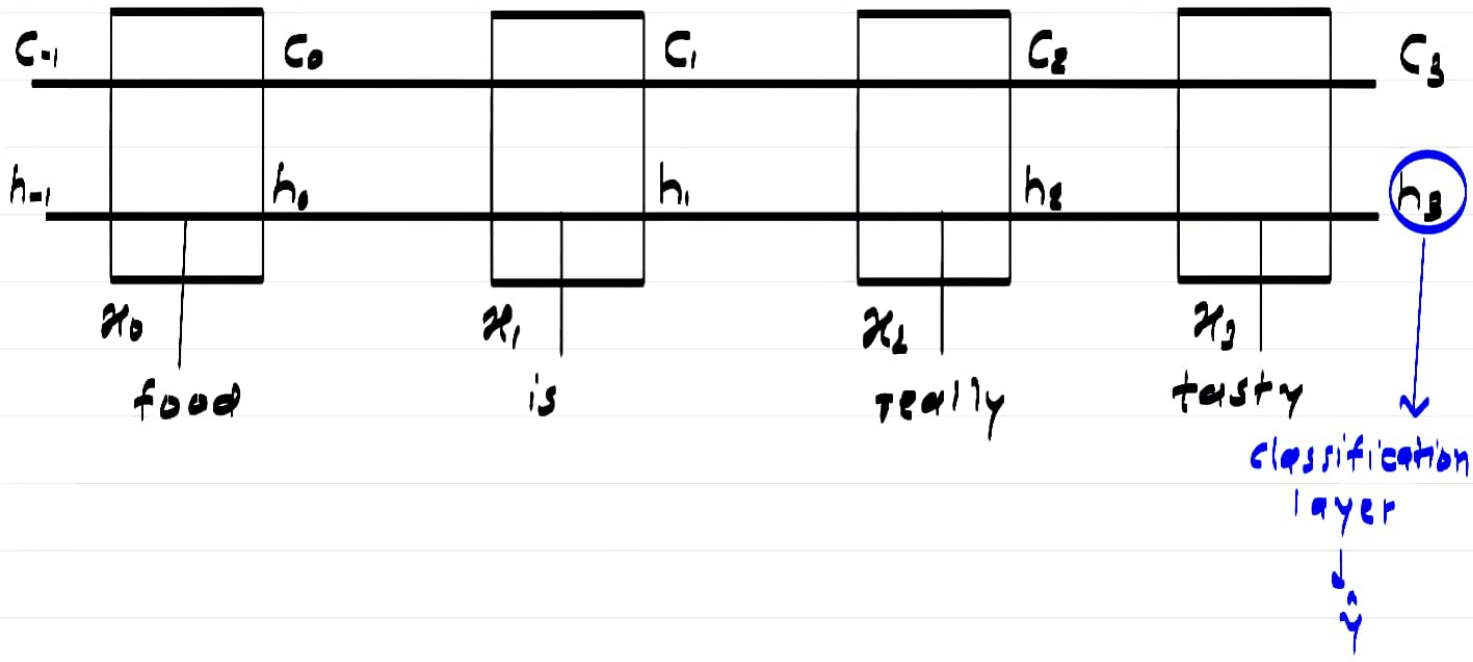
$$C_t = (f_t \times C_{t-1}) + (i_t \times \tilde{C}_t)$$

$$o_t = \sigma[w_o \times [h_{t-1} + x_t] + b_o]$$

$$h_t = o_t \times \tanh(C_t)$$

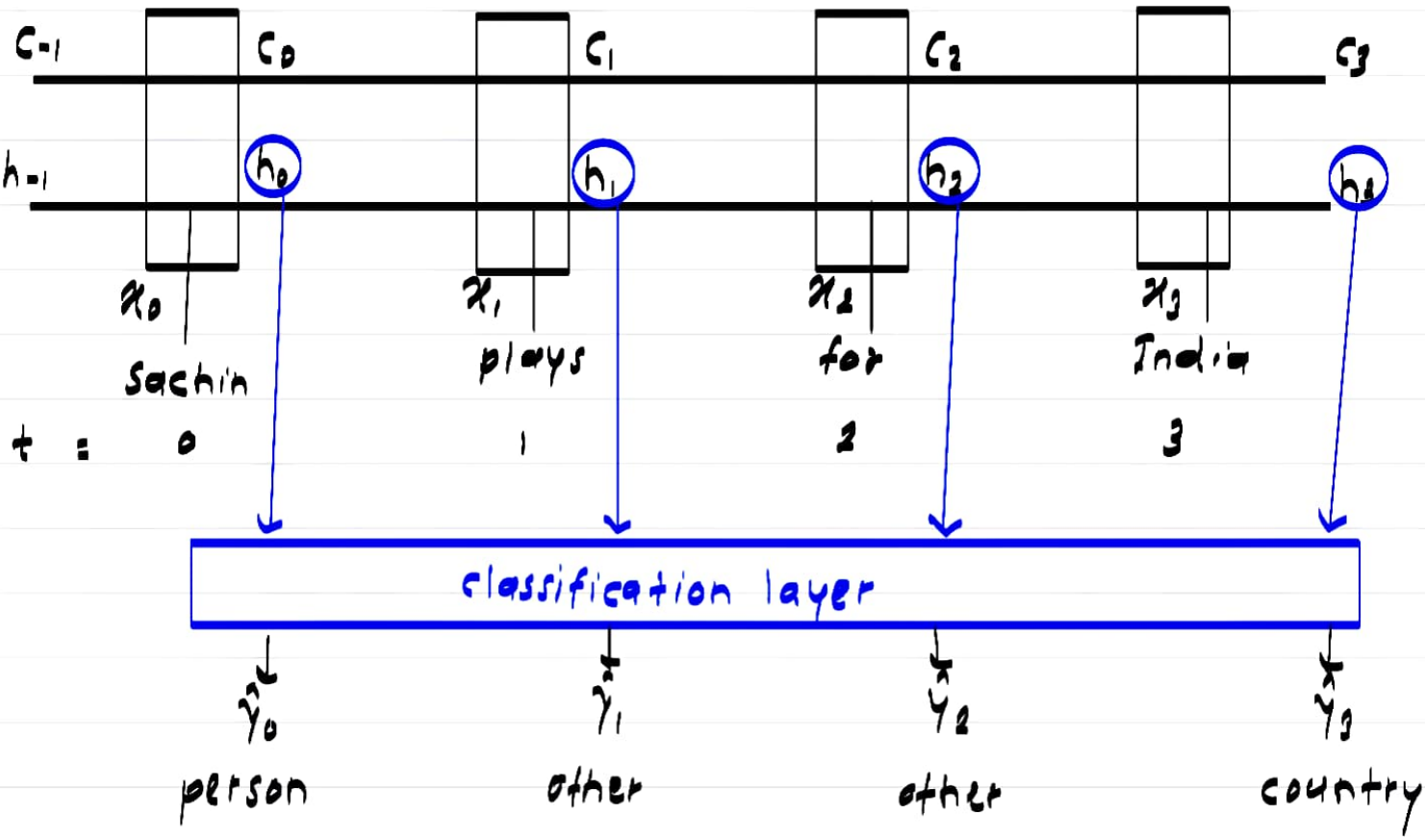
- LSTM solved the long sequence issue of RNN
- LSTM also solved problem of vanishing gradients which is very common in RNN for long sequences.

LSTM for Text classification:



- LSTM also has bidirectional variant which captures complete left to right and right to left context

LSTM for token classification:



LSTM is also limited - it works well with upto 200-250 number of tokens and gives poor performance if sequence length increases further.

Sequence to sequence tasks in NLP:

- In this type of tasks, input and output both are natural language sequences

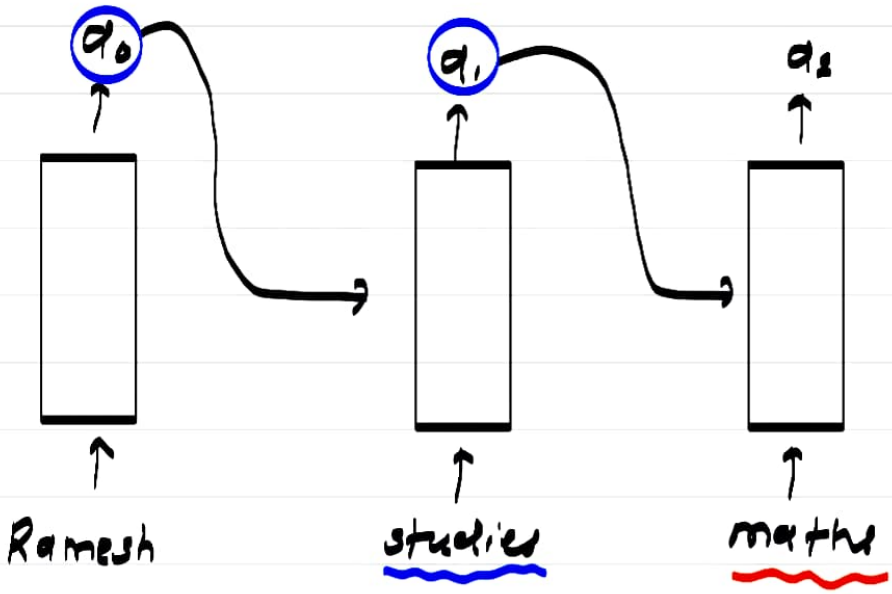
eg. QnA systems
Language translation
Speech recognition

Language translation:

- text / token classification architecture won't work for language translation / seq-seq task

English text: Ramesh studied maths.

Hindi text : Ramesh Ganit padhta hai.
रमेश गणित पढ़ता है



text / token classification architectures are not suitable for seq-seq tasks